Width is a second floor, and it is graded: the cheap end dies slowly

N=48, degree 5, 4 colours, leak-free | one knob, d_model, at L=12 attention-only | H 51223007 + I 51246202, 9 cells, 3 seeds

THE LADDER IS NOW COMPLETE, AND IT IS A RAMP

Median rho_50 across the four widths: d128 CENSORED d160 0.773 d192 0.754 d256 0.605. Every rung from d160 up crosses on 3 of 3 seeds and escapes finding 44's 1.64 plateau on 3 of 3. d128 escapes on 1 of 3 and is CENSORED -- no bin of the 14-point grid reaches solve rate 0.5. Nothing collapses between d256 and d160: the model gets worse smoothly, one rung at a time, and then stops training altogether.

WHERE THE MOVEMENT ACTUALLY IS, AGAINST THE FROZEN BAR

Consecutive gaps in units of the frozen seed-spread bar s* = 0.1484: 160->192 0.13 s*, 192->256 1.00 s*. Only the top step is separable. d160 and d192 are 0.019 apart, which is noise at this bar, so the whole 64-unit stretch from d160 to d192 -- 563,904 parameters, 42% of the smaller rung -- buys nothing a seed could not. Everything that is real lives between d192 and d256.

WHAT STAGE H COULD NOT ANSWER, AND WHY THAT WAS THE RIGHT ANSWER

H's registered gate says a rung that does not train cannot be read as a rung that cannot represent the task. d128 came in at 1 of 3 escapes, the gate fired, and H's (D)/(W) call returned INCONCLUSIVE rather than 'width is the floor'. The answer on this page instead comes from the two rungs that DID train 3 of 3 and still landed in the 0.72-0.80 barrier band: d160 at 0.773 and d192 at 0.754, both above the free line 0.74, every seed past the plateau.

WHAT DID NOT MOVE: THE STANDING MINIMUM

Neither stage fired its standing-minimum rule (3 of 3 crossing AND median rho_50 at or below 0.74). The minimum is still stage E's attention-only L=12 d256, 3,331,642 trainable at rho_50 0.605, and the best solver on this eval at any size is still the complete model at L=12. Nine GPU-cells bought a shape, not a record, and the shape is the result.

rho_50 = realized hint density at which solve rate crosses 0.5 on a 14-point x 400-instance grid, LOWER IS BETTER, reported CENSORED when no bin crosses. mean_solve = unweighted mean over the 14 bins, which never censors. s* = 0.1484 / 0.2918 was frozen before any of these cells ran. The 'free' line 0.74 is the hub (full L=6 d256, rho_50 0.592, 4,899,898 trainable) plus one s*. Commit cd5e731.

What the model actually sees

Every token is one categorical id into a 58-symbol vocabulary (N=48, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING
	BEGIN_ROUND SOLVED UNRESOLVED
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3
	the q colours (token name COLOR_n, prose 'colour')
10-57	N0 N1 ... N47
	one id per node, no structure in the id

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*

SOLVED

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 401 TOKENS

HINTS	BEGIN_HINTS N0 COLOR_0 N1 COLOR_2 N2 COLOR_3 N10 COLOR_1 N15 COLOR_1 N17 COLOR_3 ... (37 tokens)
GRAPH	BEGIN_GRAPH N0 N16 N8 N38 N34 N17 N1 N13 N27 N34 N28 N10 ... (289 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 N34 COLOR_1 N44 ... (75 tokens)

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 13 rounds, 30 deductions,
60 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000018, FALL-colouring ensemble (every node sees all q-1 other colours, so no node is permanently undeducible).

One real reasoning trace

18 hints of 48 nodes, then 13 rounds of simultaneous deduction. Hints: N0=COLOR_0 N1=COLOR_2 N2=COLOR_3 N10=COLOR_1 N15=COLOR_1 N17=COLOR_3 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND        N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 ...
  6 deduced
BEGIN_ROUND        N42 COLOR_3 N45 COLOR_1
  2 deduced
BEGIN_ROUND        N13 COLOR_0 N20 COLOR_0 N37 COLOR_2
  3 deduced
BEGIN_ROUND        N21 COLOR_2 N24 COLOR_3 N31 COLOR_2
  3 deduced
... 9 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N9, neighbours N25 N10 N29 N31 N18

their colours known before this round:

```
[0, 1, 2]  (3 = q-1 of the 4)

(3 known neighbours, 3 DISTINCT colours)

still unknown: N25 N31
```

only colour left: 3

```
so the trace emits  N9 COLOR_3
```

The model is never shown this reasoning -- only the node/colour pair.

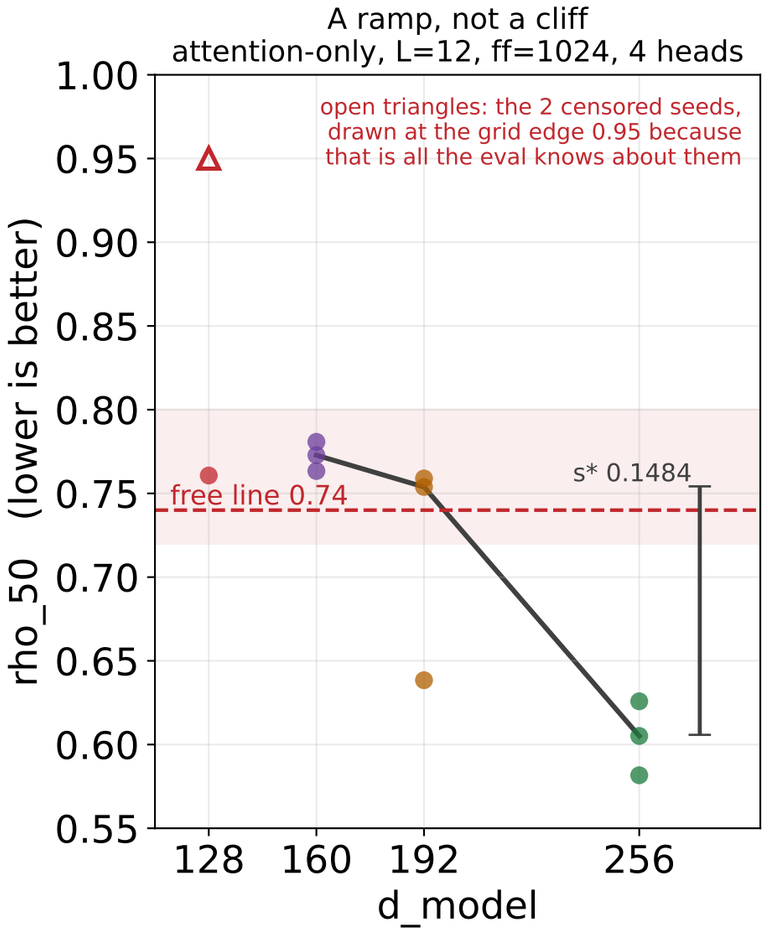
A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

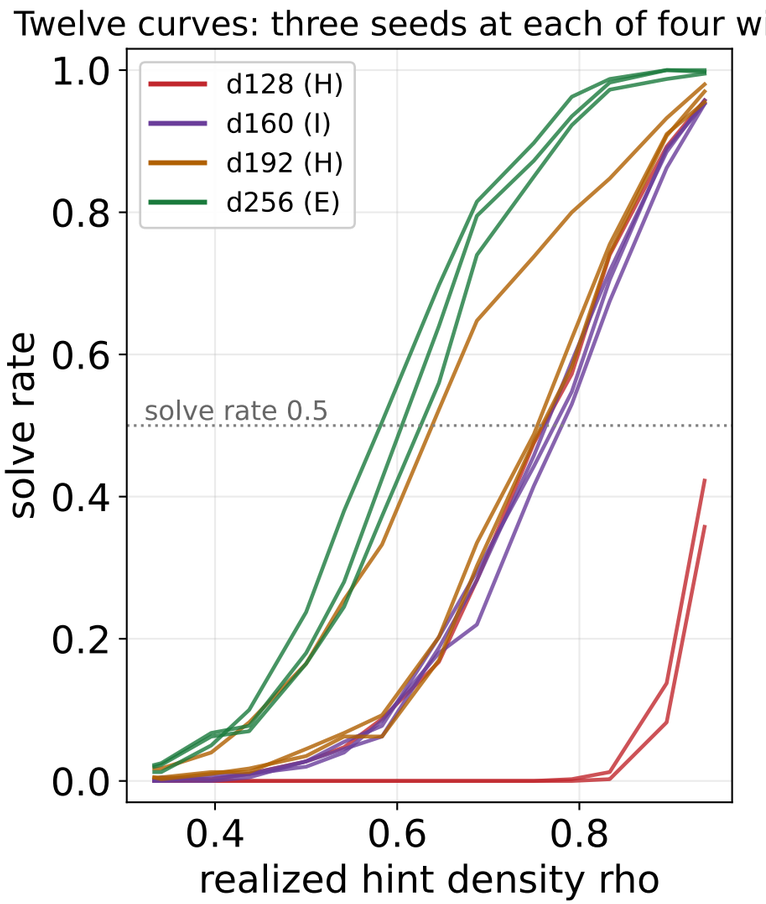
N=48, degree 5, 4 colours, --shuffle-adj --fall: the task every cell in this deck is trained and scored on. Stages H and I change nothing about it -- only d_model moves, from 256 down to 128, at a fixed depth of 12 layers and a fixed 240k-instance pool.

T=13 rounds, the 90th percentile of 40 draws from this config (range 1-16) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

The complete width ladder at L=12, and where the movement is

Left: every seed, not just the median. Right: the curves rho_50 is read off, identical 14-point x 400-instance grid for all nine cells





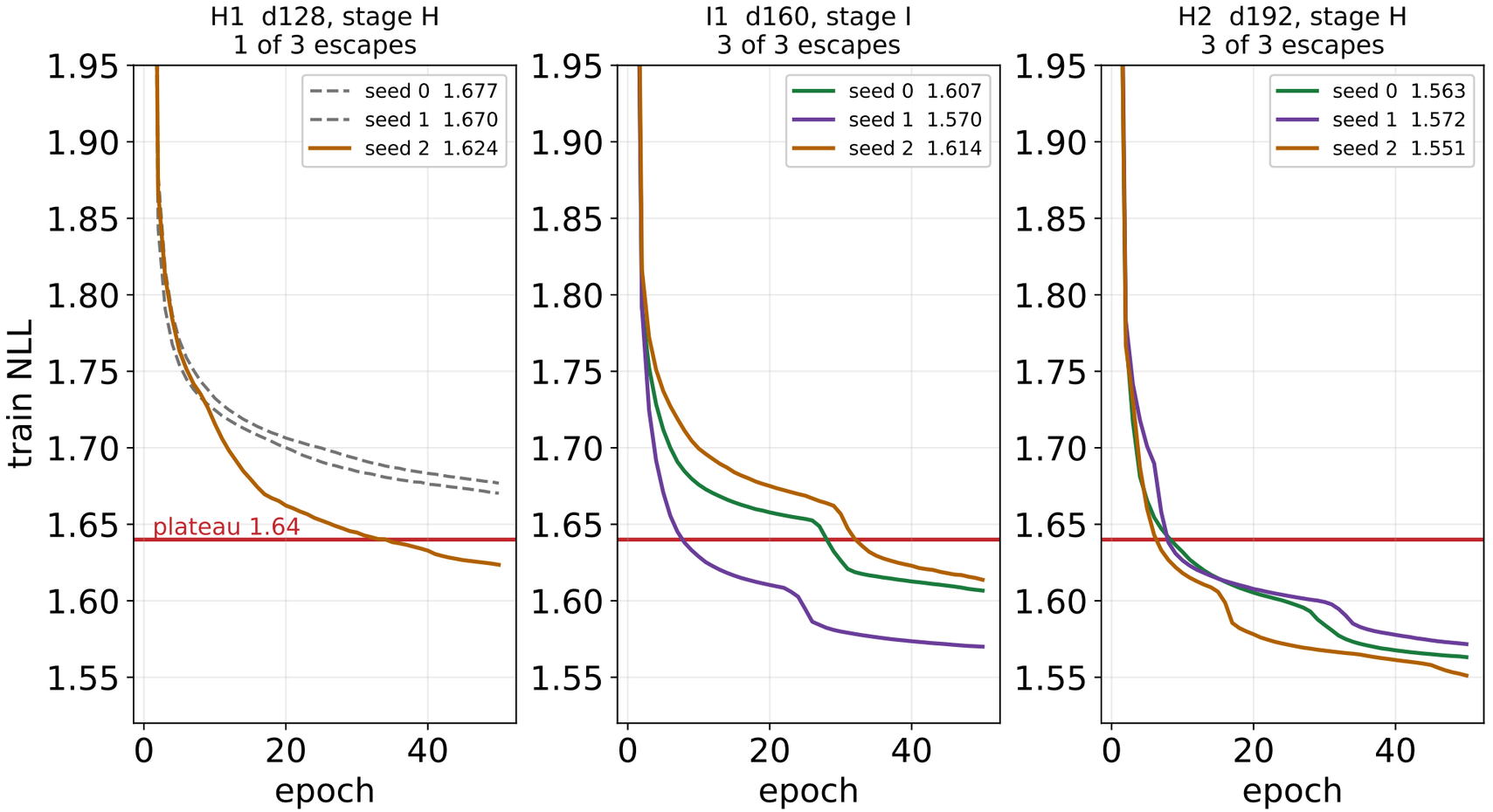
	d128 (H1)	d160 (I1)	d192 (H2)	d256 (E2)
rho_50	CENSORED	0.773	0.754	0.605
crossing	1/3	3/3	3/3	3/3
escapes	1/3	3/3	3/3	3/3
mean_solve	0.0411	0.3000	0.3220	0.5214
spread	0.2729	0.0164	0.1450	0.0480
token acc	0.4607	0.4818	0.4851	0.4926
train NLL	1.6703	1.6067	1.5632	1.5372
trainable	0.88M	1.35M	1.91M	3.33M

4*d^2 + 8*d per layer, 630*d + 58 fixed: every count above is predicted to the parameter by that formula, and asserted inside each sbatch before training starts.

Read the left panel as a ramp with a wall at the bottom. Consecutive medians: d160->d192 +0.019 (0.13 s*), d192->d256 +0.149 (1.00 s*) -- one separable step out of the two that can be read at all, and the d128 step cannot be read because that rung is censored. A censored seed's rho_50 is somewhere above the grid edge and plotting it as a number would invent data, so it is a triangle. Note the right panel: the censored d128 seeds are not shifted versions of the others, they are flat along the bottom until the very top of the grid, which is what the plateau looks like in solve space. Commit cd5e731.

Nine cells, one plateau line: which widths ever start learning

Training NLL per epoch, identical schedule and 240k pool everywhere; the red line is finding 44's plateau, the thing a cell has to leave




```
STAGE I's REGISTERED INTERVAL
committed c5a1009, before any I cell landed

  predicted final train NLL (median of 3)
      inside [1.600, 1.620]
  observed   1.6067  1.5700  1.6137
  median     1.6067   -> INSIDE

  It was an INTERVAL, not a direction: too
  low and d160 is not between its brackets,
  too high and it has not trained. Both ends
  were falsifiers and neither fired.

SEED SPREAD
  d160 mean_solve spread 0.0164,
  the 14th tightest of 31 three-seed
  rungs in the grid. Neighbours: d192 0.1450,
  d256 0.0480.
```

d128's three seeds end at 1.6768, 1.6703 and 1.6235: one crosses the line, two sit on it and are still descending at about a thousandth of a nat per epoch when the budget runs out. The registered gate therefore refuses to read this rung as a capacity statement, and the deck does not either. Stage G is the nearest evidence and it cuts both ways: at L=6 it doubled the budget to 750k steps and rescued nothing at d128, while at d256 it turned one censored seed into a solver -- so a stalled cell CAN be a budget effect, and at this width it has not been one. d160 and d192 escape on every seed, which is what makes them the rungs the (D)/(W) call is allowed to use. Commit cd5e731.

Predictions registered before the cells landed, and how they scored

Four of them, from two scripts committed while their arrays were still running; none was adjusted afterwards

H: TWO PREDICTIONS FOR d128's ESCAPES, AND THEY DISAGREED

The sbatch header said >=2 of 3 (depth has monotonically improved escapes). escape_timing.py's Q3, extrapolating each cell's terminal slope, said 0 of 3 (median 87 epochs short at the terminal slope). Observed: 1 of 3. Q3 was the closer of the two and the header was wrong -- the opposite way round from stage G, where the header won. Split one-all across two stages: neither source is reliable on escape counts, so both keep being registered.

I: THE PARAMETER COUNT, ASSERTED BEFORE TRAINING STARTED

4*d^2 + 8*d per layer plus 630*d + 58 fixed predicts 1,345,018 trainable at d160, and the sbatch aborts if the model it builds disagrees. Observed 1,345,018. The same assertion held at d128 (879,418) and d192 (1,908,922). It is a small prediction and it is the one that makes 'd160 is 2.48x smaller than the standing minimum' a measurement rather than an estimate.

I: THE TRAIN-NLL INTERVAL, AND WHAT IT WAS ANCHORED ON

Predicted median final train NLL in [1.600, 1.620] by interpolating between the neighbours AS THEY STOOD AT REGISTRATION -- and the scorer records that state so it cannot be reread favourably later: d128 had 2 of 3 cells done, d192 only 1 of 3, and that one was at epoch 40 of 50, still descending ~0.0007/epoch. The interpolation was anchored on partial data, the scorer's bracket control printed exactly that when the array closed, and the prediction still hit at 1.6067.

I: THE VERDICT THE LADDER WAS BUILT TO DECIDE

Registered as a two-way call: a CLIFF if d160 behaved like d128 (fewer than 2 of 3 escapes), GRADED if it trained and landed between its neighbours. It trained on 3 of 3 and came in at 0.773, between d128's censored and d192's 0.754 in every metric except rho_50 itself, where it is 0.019 the wrong side of d192 -- inside s*, so not a reversal. GRADED. The trainability bracket closes to [128, 160].

What a registered prediction buys here is narrow and worth stating plainly: it stops the deck choosing which of two numbers to call the result after seeing both. It does not make a 9-cell grid a large sample, and three of these four were about the ARM's behaviour, not about the metric of record. Commit cd5e731.

What H and I settle, what they leave open, and what is running tonight

One arm closed, one bracket narrowed to a gap not worth a job, and the next question already queued

SETTLED: DEPTH DOES NOT MAKE WIDTH FREE

Stage E's saving was real but it does not compose. At L=12 the attention-only model holds rho_50 0.605 at d256 and loses the free line as soon as it is narrowed: d192 0.754, d160 0.773, both on 3 of 3 trained seeds, both inside the 0.72-0.80 barrier band. Width is a second floor and it sits between d192 and d256. The breaking point is the result: this programme wants the smallest architecture that still deduces, not the highest solve rate.

SETTLED: TOKEN ACCURACY STILL EXPLAINS ALL THREE NEW RUNGS

The sufficiency fit was refit over 21 depth-ladder cells with H and I included: 21/21 well-determined held-out rungs (46 cells) sit on the depth ladder's own accuracy-to-solve curve, 20 of those cells discriminating, worst miss 0.0222 against the leave-one-out bar 0.0423. The new rungs: L=12 h=4 no_mlp d128 +0.0057, L=12 h=4 no_mlp d160 -0.0122, L=12 h=4 no_mlp d192 +0.0076. Narrowing does not open a deduction-specific failure mode; it slides the cell down one axis.

NOT SETTLED, AND DELIBERATELY NOT BEING CHASED

Whether d128 can ever learn this. It is 1 of 3 at 375k steps, and stage G's doubled budget rescued no d128 seed at L=6, so the honest label stays 'did not train'. The trainability bracket is [128, 160]; the next cell on it would cost 3 GPU-cells to split a 32-unit gap that no separable rho_50 step lives in -- d160 to d192 is already only 0.13 s*. Below d192 there is nothing left to measure, so the grid stops here.

RUNNING: STAGE J, 51280737, SHAPE AT MATCHED BUDGET

Everything separable lives between d192 and d256, and nothing in this grid sits above L=12. Stage J asks both at once, matched to 2.6% in parameters: J1 attention-only L=12 d224 (2,571,130) against J2 attention-only L=16 d192 (2,504,890), three seeds each, 6 cells, about 30 GPU-hours. Headline prediction J0: |rho_50(J1) - rho_50(J2)| <= s*, i.e. shape is not a free parameter, with a separable difference as the falsifier, named DEPTH or WIDTH in advance.

Cost for the record: 9 cells, about 37 GPU-hours, to turn 'the attention-only model is cheap' into 'the attention-only model is cheap at d256 and nowhere narrower'. No improvement is claimed: the standing minimum is unchanged at 3,331,642 trainable, and the same-width-no-depth control (attention-only L=6 d128, 480,058 trainable) is still censored at 0 of 3 escapes. Commit cd5e731.